Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
11.	(a)	(i)		+3		1		1		
		(ii)		$2FeAsS + 5O_2 \rightarrow As_2O_3 + Fe_2O_3 + 2SO_2$ formulae (1) balancing (1) – only awarded if **all** formulae are correct award (1) for correctly balanced equation including FeO instead of Fe_2O_3 $2FeAsS + 4\frac{1}{2}O_2 \rightarrow As_2O_3 + 2FeO + 2SO_2$		2		2	1	
	(b)	(i)		moles As_2O_3 in 100 cm^3 = $\frac{2.06}{197.8}$ = 0.0104 (1) moles As_2O_3 in 1 dm^3 = 0.0104 × 10 = 0.104 (1) concentration H_3AsO_3 = 0.104 × 2 = 0.208 mol dm^{-3} (1) accept alternative method		2	1	3	2	
		(ii)		$[H^+] = 10^{-5.11} = 7.76 \times 10^{-6}$ mol dm^{-3}		1		1	1	
	(c)			pyramidal (1) contains three bonding pairs and one lone pair of electrons (1) <u>electron pairs</u> arrange themselves around the central atom as far as possible from each other so that the <u>repulsion between them is at a minimum</u> / lp – bp repulsion > bp – bp repulsion (1) first two marks can be obtained from suitable diagram		1	2	3		

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	(d)			$n = \frac{pV}{RT}$ (1) $n = \frac{1.01 \times 10^5 \times 39 \times 10^{-6}}{8.31 \times 360}$ (1) n = 1.317×10^{-3} mol (1) $M_r = \frac{0.181}{1.317 \times 10^{-3}}$ = 137.4 M_r PCl_3 = 137.5 therefore chloride is PCl_3 (1) accept alternative method e.g. assume PCl_3 and work back to show volume of 39 cm^3	1	1 1	1	4	3	
	(e)	(i)		$(P^{35}Cl_2)^+$ accept if charge missing			1	1		
		(ii)		chlorine has isotopes ^{35}Cl and ^{37}Cl in ratio of 3:1 / 75% to 25% (1) since peak **C** due to $(P^{35}Cl_2)^+$ and peak **E** due to $(P^{37}Cl_2)^+$ height of **C** : **E** is (3 : 1) × (3 : 1) i.e 9 : 1 (1)		1	1	2		
				Question 11 total	**1**	**10**	**6**	**17**	**7**	**0**